

Customer Behavior Analysis

 Bold Marketing



Presented by

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Hello!

My name is Baibil shobhanshu. In this project I have utilise SQL queries to solve questions that were relted to pizza sales.



What is the total revenue generated by male vs. female customers?

```
SELECT
    gender, SUM(purchase_amount) AS revenue
FROM
    customer_shopping_behavior
GROUP BY gender;
```

	gender	revenue
▶	Male	157890
	Female	75191





Which customers used a discount but still spent more than the average purchase amount?

```
SELECT
    customer_id, purchase_amount
FROM
    customer_shopping_behavior
WHERE
    discount_applied = 'Yes'
    AND purchase_amount >= (SELECT
        AVG(purchase_amount)
    FROM
        customer_shopping_behavior);
```

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90

Which are the top 5 products with the highest average review rating?

```
SELECT
    item_purchased,
    ROUND(AVG(review_rating), 2) AS 'Average Product Rating'
FROM
    customer_shopping_behavior
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78





Compare the average Purchase Amounts between Standard and Express Shipping.

```
SELECT
    shipping_type, ROUND(AVG(purchase_amount), 2)
FROM
    customer_shopping_behavior
WHERE
    shipping_type IN ('Standard' , 'Express')
GROUP BY shipping_type;
```

shipping_type	Avg_Purchase_Amount
Express	60.48
Standard	58.46

Do subscribed customers spend more?
Compare average spend and total revenue
between subscribers and non-subscribers.

```
SELECT
    subscription_status,
    COUNT(customer_id) AS total_customers,
    ROUND(AVG(purchase_amount), 2) AS avg_spend,
    ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM
    customer_shopping_behavior
GROUP BY subscription_status
ORDER BY total_revenue , avg_spend DESC;
```

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

Which 5 products have the highest percentage of purchases with discounts applied?



```
SELECT
    item_purchased,
    ROUND(100.0 * SUM(CASE
        WHEN discount_applied = 'Yes' THEN 1
        ELSE 0
    END) / COUNT(*),
    2) AS discount_rate
FROM
    customer_shopping_behavior
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

```
with customer_type as (  
  SELECT customer_id, previous_purchases,  
  CASE  
    WHEN previous_purchases = 1 THEN 'New'  
    WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'  
    ELSE 'Loyal'  
  END AS customer_segment  
FROM customer_shopping_behavior)  
select customer_segment, count(*) AS "Number of Customers"  
from customer_type  
group by customer_segment;
```

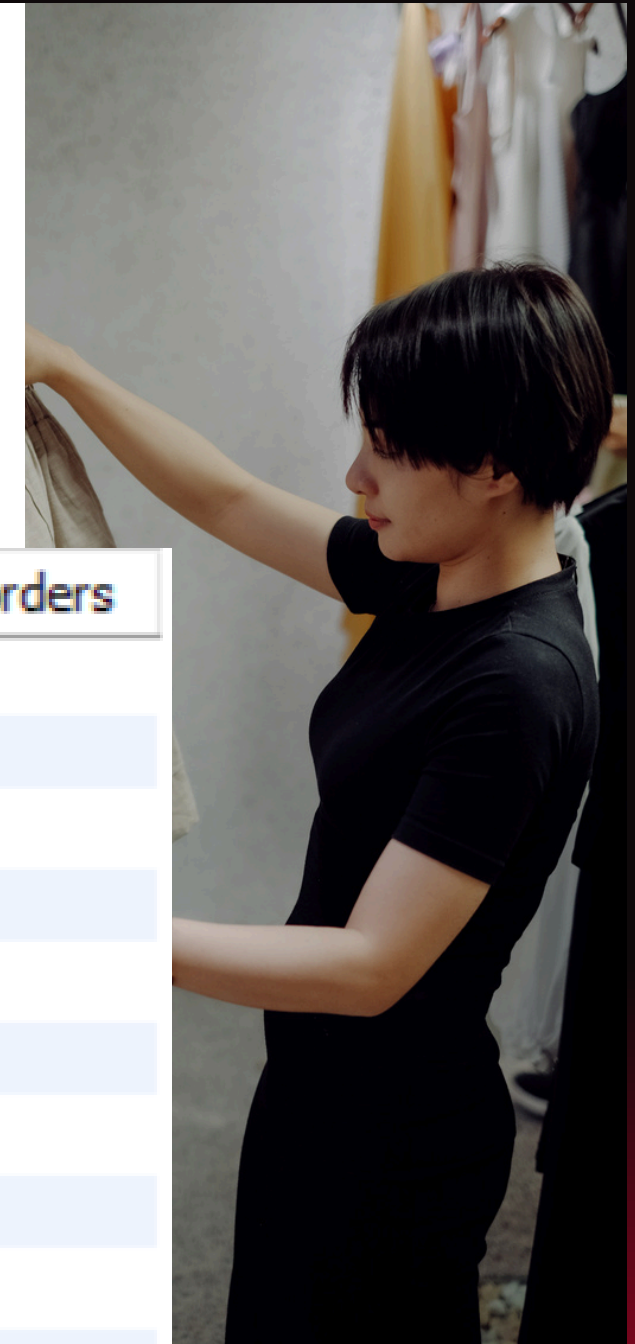


customer_segment	Number of Customers
Loyal	3116
Returning	701
New	83

What are the top 3 most purchased products within each category?

```
WITH item_counts AS (
  SELECT category,
         item_purchased,
         COUNT(customer_id) AS total_orders,
         ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank
  FROM customer_shopping_behavior
  GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;
```

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161



Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```
SELECT
    subscription_status, COUNT(customer_id) AS repeat_buyers
FROM
    customer_shopping_behavior
WHERE
    previous_purchases > 5
GROUP BY subscription_status;
```

subscription_status	repeat_buyers
Yes	958
No	2518



What is the revenue contribution of each age group?



```
SELECT
    age_group, SUM(purchase_amount) AS total_revenue
FROM
    customer_shopping_behavior
GROUP BY age_group
ORDER BY total_revenue DESC;
```

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763